Exercise 1.3

1. Solve the following simultaneous linear equations with complex coefficients for w and z.

(i) $3z + (2+i)w = 11 - i$
$(2-i)z - w = -1 + i$

(ii) $2z + (3+i)w = 9 - i$
$(3-i)z - iw = -1 + i$

2. Write the following polynomials as the product of linear factors:

(i) $z^2 + 81b^2$ (ii) $z^3 + 3z^2 + z - 5$ (iii) $4z^2 + 9b^2$
(iv) $z^3 + 3z^2 + 4z + 2$ (v) $z^3 - 7z^2 + 19z - 13$
(vi) $z^3 + 3z^2 + 19z + 17$ (vii) $z^3 - 3z^2 + 4z - 2$

3. Solve the following quadratic equations by completing the squares, where z is a complex number.

(i) $z^2 - 4z + 5 = 0$ (ii) $z^2 + 12z + 52 = 0$
(iii) $34z^2 - 6z = -1$ (iv) $z^2 - 6z + 34 = 0$
(v) $z^2 - 6z = -13$ (vi) $z^2 + 64 = 0$

Review Exercise 1

1. Select the correct option.

i. For any complex number z, $|z|$ is equal to
(a) $|\bar{z}|$ (b) $|-z|$ (c) $|-\bar{z}|$ (d) All of these

ii. If z_1 and z_2 are any two complex numbers, then
(a) $|z_1 + z_2| < |z_1| + |z_2|$ (b) $|z_1 + z_2| \leq |z_1| + |z_2|$
(c) $|z_1 - z_2| < |z_1| - |z_2|$ (d) $|z_1 + z_2| \geq |z_1| + |z_2|$

iii. If $z = 3i - 4$, then $z + \bar{z}$ = --------- (a) 8 (b) $-3i$ (c) -8 (d) $3i$-8

iv. If $a > 0$ and $b < 0$ then (a) $ab > 0$ (b) $ab < 0$ (c) $ab = 0$ (d) All of these

v. If n is an even integer, then $(i)^n$ is equal to:
(a) i (b) $-i$ (c) 1 or -1 (d) i or $-i$

vi. If z is any real number x, then its conjugate is:
(a) x (b) $-x$ (c) xi (d) $-xi$

vii. $(-1)^{\frac{-21}{2}}$ is equal to: (a) i (b) $-i$ (c) 1 (d) -1

viii. If $z = (0, 1)$ then z^2 is: (a) 1 (b) -1 (c) i (d) $-i$

ix. The product of two conjugate complex numbers is:
(a) A real number (b) An imaginary number
(c) Always zero (d) A complex number

x. Real part of $\frac{2+i}{i}$ is equal to: (a) 1 (b) 2 (c) -1 (d) $\frac{1}{2}$

xi. The plane on which complex numbers are reprsented is called:
(a) Coordinate plane (b) Complex plane
(c) Cartesian plane (d) Real plane

xii. The modulus of a complex number $z = x + iy$ is the distance of $P(x, y)$ from:
(a) x – axis (b) y – axis (c) Origin (d) (x, y)

xiii. If $z_1 = 2 + 3i$, $z_2 = 1 - i$ then $|z_1 z_2|$ = ----------
(a) $\sqrt{13}$ (b) $\sqrt{26}$ (c) $\sqrt{15}$ (d) 26

xiv. If $|x + 5i| = 3$, then x is equal to :(a) ± 4 (b)$\pm 4i$ (c)$\pm 22i$ (d)None of these

xv. Multiplicative inverse of (3,4) is:
(a) $\left(\frac{3}{25}, \frac{4}{25}\right)$ (b) $\left(\frac{3}{25}, \frac{-4}{25}\right)$ (c) $\left(\frac{-3}{25}, \frac{-4}{25}\right)$ (d) (1,0)

xvi. If $z = -3i + 4$, then $\bar{z}$ = ----------
(a) $-3i - 4$ (b)$-3i + 4$ (c) $3i + 4$ (d) $3i - 4$

xvii. Real and imaginary part of $i(3 - 2i)$ are respectively:
(a) 2 & -3 (b) 2 & 3 (c) 3 & -2 (d)-3 & -2

xviii. If $z = a + ib$, then $|z|$=: ----------
(a) $\sqrt{a - b}$ (b) $\sqrt{a^2 - b^2}$ (c) $\sqrt{a^2 + b^2}$ (d) $\sqrt{a - b}$

xix. $\left(\frac{1+i}{1-i}\right)^{12}$ = ----------
(a) -1 (b) 1 (c) i (d) $-i$

xx. The value of $i^{-7}$is: ----------
(a) 1 (b) i (c) -1 (d) $-i$

2. Simplify:
(i) $(5 - 6i) + (5i) + (7 + 6i)$ (ii) $(4 - i) + (-9 + 6i)$
(iii) $(9 + 11i) - (3 + 5i)$ (iv) $(-2 - 15i) - (-12 + 13i)$
(v) $(-3 + 2i)(3 - 8i)$ (vi) $(3 - i)(4 + 3i)(5 - 2i)$
(vii) $\frac{4-i}{6-3i}$ (viii) $\frac{8+6i}{6-2i}$

3.(a) Find each of the following:
(i) $\bar{z}$ for $z = 3 - 15i$ (ii) $\overline{z_1 - z_2}$ for $z_1 = 5 + i$ and $z_2 = -8 + 3i$
(iii) $\overline{z_1 z_2}$ for $z_1 = -10 + 5i$ and $z_2 = 5 - 10i$
(iv) $\frac{z}{\bar{z}}$ for $z = 15 - 3i$ (v) $\bar{z}_1 - \bar{z}_2$ for $z_1 = 5 + 4i$ and $z_2 = -8 + 5i$

(b) Find absolute value of the following complex numbers:
(i) $-3 - 6i$ (ii) $-6 + 3i$ (iii) $\frac{6+3i}{10+2i}$ (iv) $\frac{1+i}{1-i}$

Exercise 2.6

1. Solve the following homogeneous system of linear equations for non-trivial solutions, if possible.

(i) $\begin{aligned} x + 2y - 2z &= 0 \\ 2x + y + 5z &= 0 \\ 5x + 4y + 8z &= 0 \end{aligned}$ (ii) $\begin{aligned} x + 4y + 2z &= 0 \\ 2x + y - 3z &= 0 \\ 3x + 2y - 4z &= 0 \end{aligned}$

2. Determine the consistency of non-homogeneous system of linear equations.

(i) $\begin{aligned} x - 2y - 2z &= -1 \\ 2x + 3y + z &= 1 \\ 5x - 4y - 3z &= 1 \end{aligned}$ (ii) $\begin{aligned} x + 2y + z &= 2 \\ 2x + y + 2z &= -1 \\ 2x + 3y - z &= 9 \end{aligned}$

3. Solve the non-homogeneous system of linear equations using matrix inversion method.

(i) $\begin{aligned} x + 2y + z &= 8 \\ 2x - y + z &= 3 \\ x + y - z &= 0 \end{aligned}$ (ii) $\begin{aligned} 2x - y + 2z &= 4 \\ x + 10y - 3z &= 10 \\ -x + y + z &= -6 \end{aligned}$

4. Solve the non-homogeneous system of linear equations using Gauss elimination method.

(i) $\begin{aligned} -x + y + z &= 0 \\ x + 2y &= 5 \\ -3x + 2y - z &= -2 \end{aligned}$ (ii) $\begin{aligned} x + 2y + z &= 8 \\ 2x - y + z &= 3 \\ x + y - z &= 0 \end{aligned}$

5. Solve the non-homogeneous system of linear equations using Gauss-Jordan- method.

(i) $\begin{aligned} x - y + 4z &= 4 \\ 2x + 2y - z &= 2 \\ 3x - 2y + 3z &= -3 \end{aligned}$ (ii) $\begin{aligned} 2x + 2y - z &= 4 \\ x - 2y + z &= 2 \\ x + y &= 0 \end{aligned}$

6. Solve the non-homogeneous system of linear equations using Cramer's Rule.

(i) $\begin{aligned} x - 2y + z &= 2 \\ 2x + 2y - z &= 4 \\ x + y + 0.z &= 0 \end{aligned}$ (ii) $\begin{aligned} x - 2y + 0.z &= -4 \\ 3x + y + 0.z &= -5 \\ 2x + 0.y + z &= -1 \end{aligned}$

Review Exercise 2

1. **Select correct option.**

i. If a matrix A has m row and n column, then order of A is:

(a) $m \times n$ (b) $n \times m$ (c) mn (d) m^n

ii. Any matrix of order $m \times 1$ is called:

(a) Row matrix (b) Column matrix

(c) Square matrix (d) Zero matrix

iii. For the square matrix $A = [a_{ij}]$. If all $a_{ij} = 0, i \neq j$ and all $a_{ij} = k$ (non−zero) for $i = j$, then A is called:

(a) Diagonal matrix (b) Scalar matrix
(c) Identity matrix (d) Null matrix

iv. The matrix [7] is:

(a) Square matrix (b) Row matrix
(c) Column matrix (d) All of these

v. $(k\,ABC)^t$, where $k \neq 0$ =

(a) $kA^tB^tC^t$ (b) $kC^tB^tA^t$ (c) $\frac{1}{k}A^tB^tC^t$ (d) $\frac{1}{k}C^tB^tA^t$

vi. $\begin{vmatrix}1 & 0\\5 & 6\end{vmatrix} - \begin{vmatrix}1 & 2\\3 & 4\end{vmatrix} =$ _____

(a) 4 (b) 8 (c) −2 (d) 10

vii. If $AB = BA$, then which one is true, where A and B are square matrices:

(a) A and B are multiplicative inverses of each other
(b) One of A or B is null matrix.
(c) One of A or B is identity matrix. (d) All of these

viii. If $A = [-7]$, then $|A| =$

(a) 7 (b) −7 (c) 0 (d) Not possible

ix. Let $A = [a_{ij}]$ be a square matrix. Then cofactor of a_{ij} is equal to:

(a) M_{ij} (b) $(-1)^{i+j}M_{ij}$ (c) $(-1)^{ij}M_{ij}$ (d) $(-1)^{i+j}a_{ij}$

x. For any triangular matrix A, $|A|$ is equal to:

(a) Product of leading diagonal elements
(b) Sum of leading diagonal elements
(c) Sum of square of diagonal elements
(d) All of these

xi. A square matrix $A = [a_{ij}]$ for which all $a_{ij} = 0,\ i < j,$ then A is called:

(a) Upper triangular (b) Lower triangular
(b) Symmetric (d) Hermitian

xii. A triangular matrix is always a:

(a) Diagonal matrix (b) Scalar matrix
(c) Square matrix (d) all of these

xiii. A square matrix A is skew symmetric if:

(a) $A^t = A$ (b) $A^t = -A$ (c) $(\overline{A})^t = A$ (d) None

xiv. A square matrix A is Hermitian matrix if:

(a) $A^t = A$ (b) $A^t = -A$ (c) $(\overline{A})^t = A$ (d) $(\overline{A})^t = -A$

xv. In Skew-Hermitian matrix, elements of the main diagonal are_____ numbers.

(a) Purely real (b) Complex
(c) Either zero or purely imaginary (d) None

xvi. If $\begin{vmatrix} a & b \\ 0 & 7 \end{vmatrix} = \begin{vmatrix} 2 & 3 \\ 1 & -9 \end{vmatrix}$ then

(a) $a = -3$ (b) $a = b$ (c) $a = \frac{1}{3}$ (d) $a = \frac{-1}{3}$

xvii. The number of non-zero rows in echelon form of a matrix is called:

(a) Order of a matrix (b) Rank of a matrix

(c) Leading Column (d) Leading row

xviii. If A is any square matrix and $A = -A^t$ then A is a:

(a) Symmetric matrix (b) Skew symmetric matrix

(c) Hermitian matrix (d) Skew Hermitian matrix

xix. If A is an idempotent matrix then:

(a) $A^2 = I$ (b) $A^2 = 0$

(c) $A^2 = A$ (d) $A^2 = A^t$

xx. The cofactor A_{22} of $\begin{bmatrix} 1 & 2 & 4 \\ -1 & 2 & 5 \\ 0 & 1 & -1 \end{bmatrix}$ is:

(a) 0 (b) -1 (c) 1 (d) 2

2. Find the values of x and y if: $\begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 1 \\ 8 & 14 \end{bmatrix}$.

3. Calculate AC, BC and (A + B)C. Also verify that (A + B)C = AC + BC for

$A = \begin{bmatrix} 0 & 6 & 7 \\ -6 & 0 & 8 \\ 7 & -8 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}, C = \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}$.

4. If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ show that $A^2 - 5A + 7I_2 = 0$.

5. Find the degree or index of the nilpotent matrix $\begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}$

6. Show that matrix $\begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$ is periodic matrix of period 2.

7. Which of the following is an idempotent or an involutory matrix?

(i) $\begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (ii) $\begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$

8. Which of the following matrices are Hermitian or Skew Hermitian ?

(i) $\begin{bmatrix} 1 & 1-i & 2 \\ -1-i & 3i & i \\ -2 & i & 0 \end{bmatrix}$ (ii) $\begin{bmatrix} 4 & 1-i & 7 \\ 1+i & 6 & -i \\ 7 & i & 5 \end{bmatrix}$

Example 2. For what value of λ, the vectors $\hat{i}+2\hat{j}+3\hat{k}$, $2\hat{i}-3\hat{j}+4\hat{k}$ and $3\hat{i}+\hat{j}+\lambda\hat{k}$ are coplanar.

Solution:

Here, vectors are $\hat{i}+2\hat{j}+3\hat{k}$, $2\hat{i}-3\hat{j}+4\hat{k}$ and $3\hat{i}+\hat{j}+\lambda\hat{k}$

If the given vectors are coplanar, then $\begin{vmatrix} 1 & 2 & 3 \\ 2 & -3 & 4 \\ 3 & 1 & \lambda \end{vmatrix} = 0$

$\Rightarrow \quad 1(-3\lambda-4)-2(2\lambda-12)+3(2+9)=0$ [Expanding by R_1]

$\Rightarrow \quad -3\lambda-4-4\lambda+24+33=0$

$\Rightarrow \quad \lambda=\frac{53}{7}$

Exercise 3.6

1. Find the scalar triple product of vectors $\hat{i}+2\hat{j}+3\hat{k}, -\hat{i}-\hat{j}+\hat{k}$ and $\hat{i}+\hat{j}+\hat{k}$.

2. Compute $\left[\hat{i}+\hat{j},\ \hat{i},\ \hat{i}-\hat{j}+\hat{k}\right]$.

3. Find the volume of parallelepiped whose edges are represented by the vectors:

(i) $3\hat{i}+2\hat{j}-\hat{k}$; $\hat{i}-2\hat{j}+\hat{k}$; $\hat{i}+2\hat{j}-4\hat{k}$

(ii) $\hat{i}-2\hat{j}+3\hat{k}$; $2\hat{i}-\hat{j}-\hat{k}$; $\hat{i}+2\hat{j}-4\hat{k}$

4. Find the volume of the tetrahedron whose vertices are;
$A(2,1,8), B(3,2,9), C(2,1,4)$ and $D(3,3,10)$

5. Show that the vectors $4\hat{i}-\hat{j}+\hat{k}, 3\hat{i}-2\hat{j}-\hat{k}$ and $\hat{i}+\hat{j}+2\hat{k}$ are coplanar.

6. Find λ if the vectors $\hat{i}+\hat{j}+2\hat{k}, \lambda\hat{i}-\hat{j}+\hat{k}$ and $3\hat{i}-2\hat{j}-\hat{k}$ are coplanar.

7. Show that $\hat{i}+\hat{j}+\hat{k}$, $\hat{i}-2\hat{j}+3\hat{k}$ and $2\hat{i}-\hat{j}+3\hat{k}$ are not coplanar vectors.

Review Exercise 3

1. **Select true option.**

i. The unit vector in the direction of $\hat{r}=2\hat{i}+\hat{j}-\hat{k}$ is:

(a) $\frac{1}{\sqrt{6}}(2\hat{i}+\hat{j}-\hat{k})$ (b) $\sqrt{6}(2\hat{i}+\hat{j}-\hat{k})$

(c) $6(2\hat{i}+\hat{j}-\hat{k})$ (d) $\frac{1}{6}(2\hat{i}+\hat{j}-\hat{k})$

ii. The magnitude of $\vec{a}\times\vec{b}$ represents the _____ of a parallelogram with adjacent sides $\vec{a}$ and $\vec{b}$.

(a) opposite sides (b) diagonal (c) area (d) volume

iii. The volume of tetrahedron determined by vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ is:

(a) $[\vec{a}, \vec{b}, \vec{c}]$ (b) $\frac{1}{3}[\vec{a}, \vec{b}, \vec{c}]$ (c) $\frac{1}{6}[\vec{a}, \vec{b}, \vec{c}]$ (d) None

iv. $\vec{a} \cdot (\vec{b} \times \vec{c}) =$_____.

(a) $\vec{a}.\vec{b}.\vec{c}$ (b) $\vec{a} \times \vec{b} \times \vec{c}$ (c) $\vec{b} \cdot (\vec{a} \times \vec{c})$ (d) $(\vec{a} \times \vec{b}).\vec{c}$

v. If $\vec{a}$ and $\vec{b}$ are parallel then $\vec{a} \cdot \vec{b}$ = _____

(a) 1 (b) -1 (c) 0 (d) ab

vi. If three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are coplanar, then $[\vec{a}\ \vec{b}\ \vec{c}]$ = _____

(a) 1 (b) 0 (c) $\vec{c}$ (d) $\vec{a}$

vii. Direction cosines of the vector $\hat{i} + \hat{j} - \hat{k}$ are:

(a) $\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$ (b) $\frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

(c) $1, 1, -1$ (d) $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}$

viii. $\vec{a} \times \vec{b}$ is _____ to plane of $\vec{a}$ and $\vec{b}$:

(a) Parallel (b) Perpendicular (c) Opposite (d) None of these

ix. If $\vec{a} = \overrightarrow{P_1P_2}$, where $P_1(0, 0, 1)$ and $P_2(-3, 1, 2)$, then $|\vec{a}| =$

(a) $\sqrt{12}$ (b) $\sqrt{10}$ (c) $\sqrt{13}$ (d) $\sqrt{11}$

x. If $\vec{p} = 4\hat{i} + 6\hat{k}$ and $\vec{q} = 6\hat{i} - 4\hat{j}$ then $|\vec{p} - \vec{q}|$ is

(a) $\sqrt{8}$ (b) $2\sqrt{14}$ (c) $2\sqrt{26}$ (d) $2\sqrt{3}$

xi. For non-zero vectors $\vec{a}$ and $\vec{b}$, $\vec{a} \times \vec{b}$ is a unit vector and $|\vec{a}| = |\vec{b}| = \sqrt{2}$, then angle θ between vectors $\vec{a}$ and $\vec{b}$ is:

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{6}$ (d) $-\frac{\pi}{2}$

xii. Magnitude of a vector $\vec{a} = 3\hat{i} - \hat{j} + 2\hat{k}$ is:

(a) 13 (b) $\sqrt{12}$ (c) $\sqrt{14}$ (d) $\sqrt{11}$

xiii. The position vector of the point (1, 0, 2) is:

(a) $\hat{i} + \hat{j} + 2\hat{k}$ (b) $\hat{i} + 2\hat{j}$ (c) $2\hat{i} + 3\hat{k}$ (d) $\hat{i} + 2\hat{k}$

xiv. If O be the origin and $\overrightarrow{OP} = 2\hat{i} + 3\hat{j} - 4\hat{k}$ and $\overrightarrow{OQ} = 5\hat{i} + 4\hat{j} - 3\hat{k}$, then $\overrightarrow{PQ}$ is equal to:

(a) $7\hat{i} + 7\hat{j} - 7\hat{k}$ (b) $-3\hat{i} + \hat{j} - \hat{k}$ (c) $3\hat{i} + \hat{j} + \hat{k}$ (d) $-7\hat{i} - 7\hat{j} + 7\hat{k}$

xv. $\hat{k} \times \hat{j}$ = _____.

(a) $\hat{i}$ (b) $\hat{k}$ (c) $\hat{j}$ (d) $-\hat{i}$

xvi. If $|\vec{a} \times \vec{b}| = |\vec{a} \cdot \vec{b}|$ then the angle between $\vec{a}$ and $\vec{b}$:

(a) 0 (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{4}$ (d) π

xvii. The distance of the point $(-3,4,5)$ from the origin is:
(a) 50 (b) $5\sqrt{2}$ (c) 6 (d) None of these

xviii. The vector having, initial and terminal points as (2, 5, 0) and $(-3,7,4)$ respectively is:
(a) $-\hat{i}+12\hat{j}+4\hat{k}$ (b) $-5\hat{i}+2\hat{j}-4\hat{k}$ (c) $-5\hat{i}+2\hat{j}+4\hat{k}$ (d) $-\hat{i}+\hat{j}+\hat{k}$

xix. If $|\vec{a}|=10$, $|\vec{b}|=2$ and $\vec{a}\cdot\vec{b}=0$, then the value of $|\vec{a}\times\vec{b}|$ is:
(a) 10 (b) 16 (c) 14 (d) 20

xx. The number of vectors of unit length perpendicular to the plane of the vectors
$\vec{a}=2\hat{i}+\hat{j}+2\hat{k}$ and $\vec{b}=\hat{j}+\hat{k}$ is:
(a) One (b) Two (c) Three (d) Infinite

2. Write $\overrightarrow{PQ}$ in the form $x\hat{i}+y\hat{j}$
(i) P (4,6), Q(−4,5) (ii) $P\left(-\frac{5}{2},9\right)$, Q(10,8)

3. Find the vector PQ joining the points P(3,4,2) and Q(−2,3,4) and also find direction cosines of $\overrightarrow{PQ}$.

4. Show that P(2,3,6), Q(3,7,4) and R(4,11,−2) are collinear.

5. Show that the vectors $\frac{1}{7}\hat{i}+\hat{j}+\hat{k}$, $2\hat{i}-3\hat{j}+3\hat{k}$ and $4\hat{i}+\hat{j}-5\hat{k}$ are mutually perpendicular.

6. Find the area of parallelogram whose adjacent sides are $3\hat{i}-5\hat{j}+6\hat{k}$ and $\hat{i}+3\hat{j}-4\hat{k}$. Also find unit vector parallel to its diagonal through common initial point.

7. Prove that $[\vec{p}+\vec{q},\vec{q}+\vec{r},\vec{r}+\vec{p}]=2[\vec{p}\,\vec{q}\,\vec{r}]$.

8. For the vectors $\vec{a}=\hat{i}+2\hat{j}+\hat{k}$, $\vec{b}=3\hat{i}-\hat{j}-\hat{k}$ and $\vec{c}=\hat{i}+8\hat{j}+\hat{k}$, verify the distributive property of cross product.

Review Exercise 4

1. Select correct option.

i. A sequence is a function whose domain is set of:

(a) integers (b) rational numbers

(c) natural numbers (d) real numbers

ii. If H is the harmonic mean between x and y then H is:

(a) $\frac{2(x+y)}{xy}$ (b) $\frac{x+y}{2xy}$ (c) $\frac{2xy}{x+y}$ (d) $\frac{xy}{x+y}$

iii. If $a_n - a_{n-1} = n + 1$ and $a_4 = 14$ then a_5 = ----------:

(a) 3 (b) 5 (c) 14 (d) 20

iv. A sequence $\{a_n\}$ in which $a_{n+1} - a_n$ is the same number for all $n \in \mathbb{N}$ is called:

(a) A.P (b) G.P (c) H.P (d) None of these

v. If a_{n-1}, a_n, a_{n+1} are in A.P, then a_n is called:

(a) A.M (b) G.M (c) H.M (d) Mid-point

vi. Arithmetic mean between c and d is:

(a) $\frac{c+d}{2}$ (b) $\frac{c+d}{2cd}$ (c) $\frac{2cd}{c+d}$ (d) $\frac{2}{c+d}$

vii. The harmonic mean between $\sqrt{2}$ and $3\sqrt{2}$ is:

(a) $4\sqrt{2}$ (b) $\frac{4}{\sqrt{2}}$ (c) $\frac{3}{2}\sqrt{2}$ (d) None of these

viii. For any G.P the common ratio r is equal to:

(a) $\frac{a_n}{a_{n+1}}$ (b) $\frac{a_{n-1}}{a_n}$ (c) $\frac{a_n}{a_{n-1}}$ (d) $a_{n+1} - a_n$ for $n \in \mathbb{N}$, $n > 1$

ix. No term of a G.P is:

(a) 0 (b) 1 (c) Negative (d) Imaginary number

x. The sum of infinite geometric series is a finite number if:

(a) $|r| > 1$ (b) $|r| = 1$ (c) $|r| \geq 1$ (d) $|r| < 1$

xi. If the reciprocals of the terms of a sequence form an A.P, then it is:

(a) Harmonic sequence (b) Arithmetic sequence

(c) Reciprocal sequence (d) Series

xii. If $\frac{a^{n+1}+b^{n+1}}{a^n+b^n}$ is A.M between a & b, then n is equal to:

(a) 0 (b) -1 (c) 1 (d) $\frac{1}{2}$

xiii. If $1, x-1, 3$ are in A.P., then x = ----------

(a) 0 (b) 1 (c) 2 (d) 3

xiv. G.M between -2 and 8 is:

(a) $4i$ or $-4i$ (b) 4 or -4 (c) 16 or -16 (d) 3 or -5

xv. General term of a sequence is $(-1)^n n^2$. Its 4th term is:

(a) -4 (b) -16 (c) 16 (d) 4

xvi. If $|r| > 1$ then infinite geometric series is ----------:

(a) Convergent (b) Divergent (c) Undefined (d) both a and b

xvii. The harmonic mean of $\frac{1}{3}$ and $\frac{2}{5}$ is:

(a) $\frac{4}{11}$ (b) $\frac{3}{4}$ (c) $\frac{5}{11}$ (d) $\frac{11}{4}$

xviii. If $\frac{a-b}{b-c} = \frac{a}{b}$, then a, b and c are in:

(a) A.P (b) G.P (c) H.P (d) None of these

2. The 5th term of an arithmetic sequence is 60 and 8th term is 90. Find 12th term.

3. There are n A.Ms. between 8 and 32 such that the ratio of the third and 7th means is 3:5, find the value of n.

4. Find the sum of all the integers between 280 and 350 which are exactly divisible by 9.

5. How many terms are there in a G.P if $a = 8, a_n = \frac{1}{512}$ and $r = \frac{1}{2}$.

6. The yearly depreciation of a certain machine is 20% of the value at the beginning of the year. If the original cost of the machine is Rs. 50,000, find the value after 5 years.

7. Find r, if $S_{10} = 244S_5$ in a G.P.

8. If A.M and H.M between two numbers are 5 and $4\frac{1}{5}$ respectively. Find the numbers.

9. If G.M. and H.M. between two numbers are 15 and $5\frac{10}{13}$ respectively. Find the numbers.

10. The second term of an H.P is $\frac{1}{2}$ and the fifth term is $\frac{-1}{4}$. Find the 12th term.

$= \frac{1}{4}$ (By cancellation of terms)

Hence, $\frac{1}{1\cdot2\cdot3} + \frac{1}{2\cdot3\cdot4} + \frac{1}{3\cdot4\cdot5} + \cdots = \frac{1}{4}$.

Exercise 5.4

Find the sum of the following series:

1. $\frac{1}{1\cdot3} + \frac{1}{3\cdot5} + \frac{1}{5\cdot7} + \cdots$ to n terms.

2. $\frac{1}{1\cdot5} + \frac{1}{5\cdot9} + \frac{1}{9\cdot13} + \cdots$ to n terms.

3. $\frac{1}{1\cdot5\cdot9} + \frac{1}{5\cdot9\cdot13} + \frac{1}{9\cdot13\cdot17} + \cdots$ to infinity.

4. $\frac{1}{1\cdot4\cdot7} + \frac{1}{4\cdot7\cdot10} + \frac{1}{7\cdot10\cdot13} + \cdots$ to infinity.

Find sum of the series:

5. $\sum_{k=1}^{\infty} \frac{1}{4k^2 - 1}$

6. $\sum_{k=1}^{n} \frac{1}{9k^2 + 3k - 2}$

7. $\sum_{k=2}^{\infty} \frac{1}{k^2 - 1}$

8. $\sum_{k=2}^{n} \frac{1}{(k^2 - k)}$

Review Exercise 5

1. **Select correct option.**

i. If $a_n = 5 - 3n + 2n^2$, then a_{2n} = ----------

(a) $5 - 6n + 2n^2$ (b) $5 - 6n + 4n^2$

(c) $5 + 6n + 4n^2$ (d) $5 - 6n + 8n^2$

ii. If $a_{n-2} = 3n - 11$, then a_n = ----------

(a) $3n + 5$ (b) $3n - 5$ (c) $3n - 9$ (d) $3n - 13$

iii. $\sum_{1}^{50} n =$ ----------

(a) 1274 (b) 1275 (c) 1280 (d) 1285

iv. $\frac{1}{n(n+1)} =$ ----------

(a) $\frac{1}{n}+\frac{1}{n+1}$ (b) $\frac{1}{n}-\frac{1}{n+1}$ (c) $\frac{2}{n}+\frac{1}{n+1}$ (d) $\frac{1}{n}-\frac{3}{n+1}$

v. The nth term of $\frac{1}{2}, \frac{1}{5}, \frac{1}{8}$ ... is:

(a) $\frac{1}{3n-1}$ (b) $3n-1$ (c) $2n+1$ (d) $\frac{1}{3n+1}$

vi. $\sum n$ is equal to:

(a) $\frac{n(n+1)}{2}$ (b) $\frac{n(n+1)(2n+1)}{6}$ (c) $\frac{n^2(n+1)^2}{2}$ (d) n^2

vii. $\sum n^2$ is equal to:

(a) $\frac{n(n+1)}{2}$ (b) $\frac{n(n+1)(2n+1)}{6}$ (c) $\frac{n^2(n+1)^2}{2}$ (d) n^2

viii. $\sum n^3$ is equal to:

(a) $\frac{n(n+1)}{2}$ (b) $\frac{n(n+1)(2n+1)}{6}$ (c) $\frac{n^2(n+1)^2}{4}$ (d) $\frac{n(n+1)^2}{4}$

ix. If $S_n = (n+1)^2$, then S_{2n} is equal to:

(a) $2n+1$ (b) $4n^2+4n+1$

(c) $(2n-1)^2$ (d) Cannot be determined

x. $\sum_{n=3}^{20} n^0 =$:----------

(a) 1 (b) 19 (c) 20 (d) 18

2. Sum the following series up to n terms

$2\times1^2+4\times2^2+6\times3^2+\cdots$

3. Sum the series $\frac{1}{1\cdot4}+\frac{1}{4\cdot7}+\frac{1}{7\cdot10}+\cdots$ to infinity.

Review Exercise 6

1. Select correct answer.

i. If $n = 0$, then $\frac{(n+1)!}{n!} =$ _____

(a) 0 (b) 1 (c) n (d) ∞

ii. Probability of getting 7 in throwing a dice is:

(a) 0 (b) 1 (c) -1 (d) Not defined

iii. The factorial form of $12 \cdot 11 \cdot 10 \cdot$ is:

(a) $\frac{12!}{9!}$ (b) 12! (c) $\left(\frac{12}{9}\right)!$ (d) $(12!) \cdot (9!)$

iv. If two independent events A and B occur in p and q ways respectively, then number of ways that both events can occur is:

(a) $p + q$ ways (b) $p.q$ ways (c) $(pq)^r$ ways (d) $rp + qr$ ways

v. An arrangement of n objects according to some definite order is called:

(a) Combination (b) Permutation

(c) Factorial (d) none of these

vi. An arrangement of n objects without any order is called:

(a) Combination (b) Permutation

(c) Factorial (d) none of these

vii. The number of permutation of the letters of the word COMMITTEE is:

(a) $\binom{9}{2,\ 2,\ 2}$ (b) $\binom{6}{1,\ 2,\ 2}$ (c) $\binom{9}{2,\ 2,\ 1}$ (d) $\binom{9}{2,\ 3,\ 1}$

viii. 8.7.6 is equal to:

(a) ${}^8P_3$ (b) ${}^8C_3$ (c) ${}^8P_5$ (d) ${}^8C_5$

ix. If $r = n$, then ${}^nP_r$ is equal to:

(a) $r!$ (b) $(n - r)!$ (c) 1 (d) 0

x. The number of ways that a necklace of n beads of different colours be made is:

(a) $(n - 1)!$ (b) $\frac{n!}{2}$ (c) $\frac{n!-1}{2}$ (d) $\frac{(n-1)!}{2}$

xi. Any subset of a sample space is called:

(a) Sample space (b) an event

(c) a Trial (d) Random variable

xii. For two events A and B if $A \cap B = \emptyset$, then events A and B are called:

(a) Mutually exclusive (b) Not mutually exclusive

(c) Overlapping (d) Dependent events

xiii. When a dice is rolled and coin is tossed, all possible outcome are:

(a) 6 (b) 12 (c)18 (d)24

xiv. If two events A and B have equal chance of occurrence, then the events are:

(a) Equally likely (b) Not equally likely

(b) Dependent (d) Not mutually exclusive

xv. If E be an event of a sample space S, then:

(a) $P(E) = \frac{n(S)}{n(E)}$ (b) $0 \leq P(E) \leq 1$

(c) $0 < P(E) < 1$ (d) all of these

xvi. The probability of getting the tail in a single toss of a coin is:

(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

xvii. Three dice are rolled simultaneously, then $n(S)$ is equal to:

(a) 36 (b) 18 (c) 216 (d) 6

xviii. Two teams A and B are playing a match, the probability that team A does not lose is:

(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{1}{2}$ (d) 0

xix. If ${}^{n}C_6 = {}^{n}C_{12}$, then n equals:

(a) 18 (b) 12 (c) 6 (d) 20

2. Evaluate the following:

(i) $\frac{{}^{6}C_3 \times {}^{4}C_3}{{}^{10}C_4}$ (ii) $\frac{{}^{6}P_3 \times {}^{4}P_3}{{}^{10}P_4}$ (iii) $\sum_{r=1}^{3} {}^{3}C_r$

3. There are seven seats available in a compartment. In how many ways can seven persons be seated?

4. A room has 3 lamps. From a collection of 10 light bulbs of which 6 are no good, a person selects 3 at random and puts them in the sockets. What is the probability that he will have light?

5. If two dice are thrown simultaneously, what is the probability of obtaining a sum of 7 or a sum of 11?

6. A bag contains 8 white, 10 black and 12 red balls. 3 balls are drawn from the bag. What is the probability that the first ball is white, the second ball is black and the third ball is red, when every time the ball is replaced?

7. In how many ways can a football team of 11 players be selected out of 15 players? How many of them will include a particular player?

then show that $4x^2 - 2x - 1 = 0$.

9. If $\frac{y}{2} = \frac{1}{2}\left(\frac{4}{9}\right) + \frac{1\cdot3}{2^2\cdot2!}\left(\frac{4}{9}\right)^2 + \frac{1\cdot3\cdot5}{2^3 3!}\left(\frac{4}{9}\right)^3 + \cdots$,
then show that $5y^2 + 20y - 16 = 0$.

Review Exercise 7

1. **Select correct answer.**

i. The general term of the binomial expansion $(a + x)^n$ is where $n \in N$

(a) $\binom{n}{r} a^n x$ (b) $\binom{n}{r} a^{n-r} x^r$ (c) $\binom{n}{r} a^r x^{n-r}$ (d) $\binom{n}{r} (ax)^{n-r}$

ii. The number of terms in the expansion of $(a + b)^{2n}$ are:

(a) n (b) $2n + 1$ (c) 2^n (d) 2^{n-1}

iii. In the expansion $(a + x)^n$, the exponent of 'x':

(a) decreases from n to 0 (b) increases from 0 to n

(c) remains n every where (d) becomes 0 at the end

iv. Middle term in the expansion of $(a + b)^{2n}$ is:

(a) nth term (b)(n + 1)th term

(c) (2n + 1)th term (d) (2n − 1)th term

v. The term independent of x in the expansion of $(a + 2x)^n$ is:

(a) First term (b) Middle term

(c) Last term (d) 2nd last term

vi. The coefficient of the last term in the expansion of $(2 - x)^7$ is/are:

(a) 1 (b) −1 (c) 7 (d) −7

vii. In the expansion $\left(a + \frac{1}{2}\right)^7$, the number of middle terms is/are:

(a) one (b) two (c) three (d) four

viii. Sum of odd binomial coefficients in the expansion of $(a + x)^n$ is:

(a) 2^n (b) 2^{n-1} (c) 2^{n+1} (d) $n + 1$

ix. $\binom{n+1}{0} + \binom{n+1}{1} + \binom{n+1}{2} + \cdots + \binom{n+1}{n+1}$ is equal to:

(a) 2^n (b) 2^{n+1} (c) 2^{n-1} (d)Cannot be determined

x. The number of terms in the expansion of $(1 + x)^{\frac{1}{3}}$ is _____.

(a) $\frac{4}{3}$ (b) 4 (c) ∞ (d) 2

xi. $1 - x + x^2 - x^3 + \cdots$ is equal to:

(a) $(1 + x)^{-1}$ (b) $(1 - x)^{-1}$ (c) $(1 + x)^{-2}$ (d) $(1 - x)^{-2}$

xii. The expansion of $(1 - 2x)^{-2}$ is valid if:

(a) $|x| < 0$ (b) $|x| < \frac{1}{2}$ (c) $|x| < 2$ (d) $|x| < 1$

xiii. The middle term in the expansion of $(a+b)^n$ is $\left(\frac{n}{2}+1\right)$ th term; then n is:

(a) Odd (b) Even

(c) Prime (d) None of these

xiv. In $\left(a+\frac{1}{a}\right)^8$, the sum of the binomial coefficients is:

(a) 64 (b)128 (c)256 (d) 512

2. Prove by principle of mathematical induction $\forall n \in \mathrm{N}$ that $8 \times 10^n - 2$ is divisible by 6.

3. Prove by mathematical induction

$1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$

4. Given that the coefficient of y^2 in the expansion $\frac{1}{(1+my)^2}$ is 96, find m.

5. Given that the terms involving m^4 and higher powers may be neglected and that $\frac{1}{(a+bm)^3} - \frac{1}{(1+3m)^4} = cm^2 + dm^3$, find the values of a, b, c and d.

Review Exercise 8

1. Select correct answer.

i. If $f: A \to B$ be a function, then it is an onto function if:

(a) Range = B (b) Range $\subset$ B

(c) Image is not repeated (d) Domain $\neq$ A

ii. The graph of $y = x^6$ is symmetric to _____.

(a) $x-$axis (b) $y-$axis (c) origin (d) none

iii. An one to one function is also called _____ function:

(a) Injective (b) Surjective (c) Bijective (d) Inverse

iv. Inverse of a function exists only if it is:

(a) Injective (b) Bijective (c) Surjective (d) all of these

v. The function $f = \{(x, y) \mid y = mx + c\}, m$ and c are real numbers is:

(a) Linear (b) Quadratic (c) A circle (d) A point

vi. The function $f = \{(x, y) \mid y = ax^2 + bx + c, a \neq 0\}$ is:

(a) Linear (b) Quadratic (c) A circle (d) A point

vii. The range of $f(x) = x^2 + 3 \ \forall x \in \mathbb{R}$ is _____.

(a) $\mathbb{R}$ (b) $\mathbb{R}^-$ (c) $(3, \infty)$ (d) $[3, \infty]$

viii. Graph of $y = x^n$ is symmetric to _____ if n is an odd integer.

(a) $x-$axis (b) $y-$axis (c) origin (d) none

ix. The graph of linear function is:

(a) Circle (b) Straight line

(c) Parabola (d) Triangle

x. If f is function from A to B. Domain of f is equal to:

(a) Any subset of A (b) A $\times$ B (c) A (d) B

xi. Every function is a:

(a) Relation (b) Inverse function

(c) One to one (d) None of these

xii. If f and g are equal where $f(x) = 7x - 4$ and $g(x) = x$ then $x =$:

(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) 1 (d) $\frac{4}{3}$

xiii. Domain of the function $f(x) = \frac{3x+1}{x-1}$ is set of all:

(a) real numbers (b) rational numbers

(c) complex numbers (d) real numbers except 1

xiv. A function $f(x) = |x| - x^2$ is:

(a) odd (b) Linear

(c) Even (d) Neither even nor odd

(iii) Maximize $Z(x, y) = 41x + 38y$
Subject to: $4x + 5y \leq 26$; $8x + 5y \leq 22$; $5x + 2y \leq 10$

(iv) Maximize $Z = 4x + y$
Subject to: $x + y \leq 50$; $3x + y \leq 90$; $x - y \leq 40$

(v) Minimize $Z = 200x + 500y$
Subject to the constraints: $x + 2y \geq 10$; $3x + 4y \leq 24$

(vi) Minimize $Z = 3x + 2y$
Subject to the constraints: $2x + y \geq 6$; $x + y \geq 4$

(vii) Minimize $Z = 0.12x + 0.15y$
Subject to the constraints: $x + y \geq 5$; $2x + y \geq 6$; $x + 3y \geq 9$

2. A landlord has 50 hectares of land to grow two crops X and Y with profits per hectare are estimated as Rs 10,500 and Rs 9,000 respectively. To control weeds, a liquid herbicide no more than 800 litres, is to be used for crops X and Y at rates of 20 liters and 10 liters per hectare. How much land should be allocated to each crop so as to maximize the total profit.

3. A toy factory manufactures two types of toys A and B and sells for Rs.25 and Rs.20 respectively. The toys A and B respectively, require 20 units and 12 units daily out of 2000 available resource units. Both require a production time of 5 minutes and total working hours are 9 per day. What should be the quantity for each of the toys to maximize the selling amount?

4. At a university, Professor Abdul Sattar wishes to employ two people, Farhan and Sarfaraz, to grade papers for his classes. Farhan is a graduate student and can grade 20 papers per hour; Farhan earns \$15 per hour for grading papers. Sarfaraz is a post-doctoral associate and can grade 30 papers per hour; Sarfaraz earns \$25 per hour for grading papers. Each must be employed at least one hour a week to justify their employment. If Prof. Abdul Sattar has at least 110 papers to be graded each week, how many hours per week should he employ each person to minimize the cost?

Review Exercise 9

1. **Select correct answer.**

i. If $2x - 5 \geq 3$ then its solution set is _____, where $x \in \mathbb{Z}$.

(a) $\{2, 3, 4, \ldots\}$ (b) $\{4, 5, \ldots\}$ (c) $\{2, 1, 0, \ldots\}$ (d) $\{1, 0, -1, \ldots\}$

ii. If $4x - 3y = 8$ is the corresponding equation of an inequality $4x - 3y \leq 8$ then its x-intercept is _____.

(a) $\frac{3}{2}$ (b) $\frac{2}{3}$ (c) 2 (d) $\frac{1}{2}$

iii. $\{x | x \in \mathbb{R} \wedge x > 5\} =$ _____.

(a) $[5, \infty)$ (b) $(-\infty, 5)$ (c) $(5, \infty)$ (d) $(-\infty, 5]$

iv. Linear Programming (LP) is used to obtain _____solution.

(a) Feasible (b) Trivial (c) Optimal (d) Infeasible

v. Solution space of the linear inequality $2x + 3y \leq 6, \forall x, y \in \mathbb{R}$ includes ____.

(a) All points above the line (b) All points on and below the line

(c) All points below the line (d) All points on and above the line

vi. If $f(x, y) = 5x + 4y$ is an objective function and the corner points of the feasible region are $(5, 4), (0, 0)$, $(4, 0)$ and $(0, 6)$ then the function is maximum at:

(a) (4,0) (b) (0,6) (c) $(0, 0)$ (d) $(5, 4)$

vii. _____ are the entities whose values are to be determined from the solution of the LP problem.

(a) Objective function (b) Decision Variables

(c) Constraints (d) Opportunity cost

viii. _____ specifies the objective or goal of solving the LP problem.

(a) Objective function (b) Decision Variables

(c) Constraints (d) Opportunity cost

ix. _____ are the restrictions or limitations imposed on the LP problem:

(a) Variables (b)Costs (c) Profits (d)Constraints

x. The region of solution in LP problem is called ____.

(a) Infeasible region (b) Unbounded region

(c) Infinite region (d) Feasible region

xi. In case of an '_____' constraint, the feasible region is a straight line.

(a) Less than or equal to (b) Greater than or equal to

(c) Mixed (d) Equal to

xii. $ax + b \leq 0,\ a \neq 0,\ a, b \in \mathbb{R}$, is called _____.

(a) Non-linear inequality (b) Linear inequality

(c) Linear equality (d) Complex inequality

xiii. A point of a solution region where two of its boundary lines intersect, is called _____.

(a) Middle point (b) Origin

(c) Corner point (d) Feasible point

xiv. The solution region of an inequality restricted to the first quadrant is called _____ region.

(a) Combined (b) Unbounded

(c) Infeasible (d) Feasible

xv. A function which is to be maximized or minimized is called _____.

(a)Linear function (b) Equal function

(c) Objective function (d) Non linear function

xvi. The feasible solution which maximizes or minimizes the objective function is called the _____.

(a) Optimal solution (b) Corner solution

(c) Initial solution (d) Complex solution

2. Draw graph of the following system of linear inequalities and find feasible region.

$2x + 3y \geq 10; \quad 3x + 2y \geq 12; \quad x \geq 0, y \geq 0$

3. Find the feasible region graphically subject to the following constraints and also find its corner points.

$4x + y \geq 9; \quad 2x + 3y \leq 14; \quad x - y \leq 5; \quad x \geq 0, y \geq 0$

4. Find the maximum and minimum values of the function $f(x, y) = 4x - 3y$ subject to the constraints

$2x + 3y \geq 5; \quad 3x + 2y \leq 12; \quad x \geq 0, y \geq 0$

5. A dealer wishes to purchase some of fans and sewing machines. He has only ¥ 5760 to invest and space of at most 20 items. The costs of a fan and a sewing machine are ¥ 360 and ¥ 240 respectively. He wants profit on a fan and a sewing machine ¥ 22 and ¥ 18 respectively. Assuming that he can sell all the items that he can buy, how should he invest his money in order to maximize his profit?

Review Exercise 10

1. Select the correct option.

(i) $\cos(\alpha - \beta)$ is equal to:

(a) $\cos\alpha\cos\beta + \sin\alpha\sin\beta$ (b) $\cos\alpha\cos\beta - \sin\alpha\sin\beta$

(c) $\sin\alpha\cos\beta + \cos\alpha\sin\beta$ (d) $\sin\alpha\cos\beta - \cos\alpha\sin\beta$

(ii) Angles associated with basic angles of measure θ to a right angle or its multiple are called:

(a) Coterminal angles (b) Angles in standard positions

(c) Allied angles (d) Obtuse angles

(iii) $\sin\left(\frac{\pi}{2} + \theta\right)$ is equal to:

(a) $\cos\theta$ (b) $\sin\theta$ (c) $-\cos\theta$ (d) $-\sin\theta$

(iv) $\cos(\pi - \theta)$ is equal to:

(a) $\sin\theta$ (b) $\cos\theta$ (c) $-\sin\theta$ (d) $-\cos\theta$

(v) $\tan(\pi + \theta)$ is equal to:

(a) $\tan\theta$ (b) $-\cot\theta$ (c) $-\tan\theta$ (d) $\cot\theta$

(vi) $\cos(2\pi + \theta)$ is equal to:

(a) $\sin\theta$ (b) $\cos\theta$ (c) $-\sin\theta$ (d) $-\cos\theta$

(vii) $\sin 540°$ is equal to:

(a) 1 (b) 0 (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{\sqrt{3}}{2}$

(viii) $\sin(-300°)$ is equal to:

(a) 1 (b) $\frac{\sqrt{3}}{2}$ (c) 0 (d) -1

(ix) If α, β and γ are the angles of Δ ABC, then $\sin(\alpha + \beta)$ is equal to:

(a) $\sin\gamma$ (b) $-\sin\gamma$ (c) $\cos\gamma$ (d) $-\cos\gamma$

(x) $\cos 2\alpha$ is equal to:

(a) $\cos^2\alpha - \sin^2\alpha$ (b) $2\cos^2\alpha - 1$ (c) $1 - 2\sin^2\alpha$ (d) all of these

(xi) $\sin\frac{\alpha}{2}$ is equal to:

(a) $\pm\sqrt{\frac{1+\sin\alpha}{2}}$ (b) $\pm\sqrt{\frac{1-\cos\alpha}{2}}$ (c) $\pm\sqrt{\frac{1+\cos\alpha}{2}}$ (d) $\pm\sqrt{\frac{1-\sin\alpha}{2}}$

(xii) $\cos 3\alpha$ is equal to: (a) $3\cos\alpha - 4\cos\alpha$ (b) $3\cos^3\alpha + 4\cos\alpha$

(c) $4\cos^3\alpha - 3\cos\alpha$ (d) $4\cos^3\alpha + 4\cos\alpha$

(xiii) $\cos\alpha - \cos\beta =$ (a) $2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$ (b) $2\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$

(c) $2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$ (d) $-2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$

2. Find r_1, r_2 and r_3, if the measures of sides of triangle ABC are

(i) $a = 23cm,$ $b = 24cm,$ $c = 28cm$

(ii) $a = 24.4cm,$ $b = 34.8cm,$ $c = 42.5cm$

(iii) $a = 75cm,$ $b = 62cm,$ $c = 53cm$

3. If $a = b = c$, then prove that

$$r_1 : R : r = 3 : 2 : 1$$

4. Show that:(i) $r_1 r_2 r_3 = rs^2$ (ii) $r = 4R\sin\frac{\alpha}{2}\sin\frac{\beta}{2}\sin\frac{\gamma}{2}$ (iii)$r_1 = \frac{a.\cos\frac{\beta}{2}\cos\frac{\gamma}{2}}{\cos\frac{\alpha}{2}}$

5. Prove that $\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{a^2+b^2+c^2}{\blacktriangle^2}$

6. Show that $\frac{1}{2rR} = \frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca}$

Review Exercise 11

1. **Select correct answer.**

i. From the top of a cliff 80m high the angle of depression of a boat is α. If the distance between the boat and foot of cliff is $80\sqrt{3}$ m, then angle α is: (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{6}$ (c) $\frac{\pi}{3}$ (d) $\frac{3\pi}{4}$

ii. To solve an oblique triangle, we use:

(a) law of sines (b) Laws of cosines (c) Laws of tangents (d) All of these

iii. A circle which touches all the sides of a triangle is called _____.

(a) circum-circle (b) in-circle (c) ex-circle (d) tri-circle

iv. In a triangle ABC, if $\beta = 90°$, then $b^2 = c^2 + a^2 - 2ac\cos\beta$ becomes:

(a) Law of sines (b) Law of tangents

(c) Pythagoras theorem (d) None of these

v. In any triangle ABC, $\sqrt{\frac{(s-a)(s-c)}{ac}}$ is equal to:

(a) $\sin\frac{\alpha}{2}$ (b) $\cos\frac{\alpha}{2}$ (c) $\sin\frac{\beta}{2}$ (d) $\sin\frac{\gamma}{2}$

vi. In any triangle ABC, $\cos\frac{\gamma}{2}$ is equal to:

(a) $\sqrt{\frac{s(s-a)}{ab}}$ (b) $\sqrt{\frac{s(s-b)}{ac}}$ (c) $\sqrt{\frac{s(s-a)}{bc}}$ (d) $\sqrt{\frac{s(s-c)}{ab}}$

vii. In any triangle ABC, with usual notations, s is equal to:

(a) $a + b + c$ (b) $\frac{a+b+c}{2}$ (c) $\frac{a+b+c}{3}$ (d) $\frac{abc}{2}$

viii. $\sqrt{\frac{s(s-b)}{(s-a)(s-c)}} = \cdots$ (a) $\sin\frac{\beta}{2}$ (b) $\cos\frac{\beta}{2}$ (c) $\tan\frac{\beta}{2}$ (d) $\cot\frac{\beta}{2}$

ix. In any triangle ABC, $\sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$ is equal to:

(a) $\sin\frac{\gamma}{2}$ (b) $\cos\frac{\gamma}{2}$ (c) $\tan\frac{\gamma}{2}$ (d) $\cot\frac{\gamma}{2}$

x. We can solve an oblique triangle, if:

(a) One side and two angles are known

(b) Three sides are known

(c) Two sides and their included angles are known

(d) All of these

xi. In ΔABC, inradius = ----------

(a) $\frac{\blacktriangle}{s-a}$ (b) $\frac{\blacktriangle}{s-b}$ (c) $\frac{\blacktriangle}{s-c}$ (d) $\frac{\blacktriangle}{s}$

xii. In any triangle ABC, Area of triangle is:

(a) $bc\sin\alpha$ (b) $\frac{1}{2}ca\sin\alpha$ (c) $\frac{1}{2}ab\sin\gamma$ (d) $\frac{1}{2}ab\sin\beta$

xiii. Hero's formula is:

(a) $\blacktriangle = s(s-a)(s-b)(s-c)$ (b) $\blacktriangle = \sqrt{(s-a)(s-b)(s-c)}$

(c) $\blacktriangle = \sqrt{s(s-a)(s-b)(s-c)}$ (d) $\blacktriangle = \frac{a+b+c}{2}$

xiv. The circle passing through the three vertices of a triangle is called:

(a) Circum-circle (b) In-circle (c) Ex-centre (d) Escribed circle

xv. The point of intersection of the right bisectors of the sides of a triangle is called:

(a) Circum-centre (b)In-centre (c)Escribed centre (d) Ortho-centre

xvi. Radius of the circle which passes through all the vertices of a triangle is:

(a) Circum-radius (b) In-radius (c) e-Radius (d) Diameter

xvii. In any triangle ABC, with usual notations, abc =

(a) R (b) Rs (c) 4R▲ (d) $\frac{\blacktriangle}{s}$

xviii. The point of intersection of the internal bisectors of angles of a triangle is:

(a) In-centre (b) e-centre (c) Circum-centre (d) Ex-centre

xix. In any equilateral triangle ABC, with usual notations, $r:R:r_1$

(a) 1:2:3 (b) 3:2:1 (c) 1:3:2 (d) 1:1:1

xx. Circum radius of ΔABC is:

(a) $\frac{\blacktriangle}{S}$ (b) $\frac{\blacktriangle}{S-b}$ (c) $\frac{\blacktriangle}{S-a}$ (d) $\frac{abc}{4\blacktriangle}$

2. Prove that:

(i) $r_2 = \frac{b\cos\frac{\alpha}{2}\cos\frac{\gamma}{2}}{\cos\frac{\beta}{2}}$ (ii) $r_1r_2 + r_2r_3 + r_3r_1 = s^2$

Review Exercise 12

1. Select correct answer.

i. If $y = \text{Sin}\, x$, then domain is:

(a) $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$ (b) $0 \le x \le \pi$ (c) $[0, x], x \ne \frac{\pi}{2}$ (d) None of these

ii. If $y = \text{Cos}\, x$, then range is:

(a) $-1 \le y \le 1$ (b) R (c) $y \le -1$ or $y \ge 1$ (d) $y \le -1$ or $y \ge 1$

iii. If $y = \text{Tan}^{-1}\, x$, then domain is:

(a) $-1 \le x \le 1$ (b) R (c) $x \ge -1$ or $x \le 1$ (d) $x \le -1$ or $x \ge 1$

iv. If $y = \text{Sec}^{-1}\, x$, then range is:

(a) $0 \le y \le \pi,\ y \ne \frac{\pi}{2}$ (b) $-\frac{\pi}{2} \le y \le \frac{\pi}{2},\ y \ne 0$

(c) $0 < y < \pi$ (d) $-\frac{\pi}{2} < y < \frac{\pi}{2}$

v. Inverse function exists if and only if the function is:

(a) bijective (b) (1-1) function

(c) onto function (d) into function

vi. $\text{Cot}^{-1}\, x =$ .....

(a) $\frac{\pi}{2} - \tan^{-1} x$ (b) $\frac{\pi}{2} - \cot^{-1} x$ (c) $\frac{\pi}{2} + \tan^{-1} x$ (d) $\frac{\pi}{2} + \cot^{-1} x$

vii. $\text{Sec}\left[\text{Sin}^{-1}\left(-\frac{1}{2}\right)\right] =$ .....

(a) $\frac{2}{\sqrt{3}}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{1}{2}$ (d) $-\frac{2}{\sqrt{3}}$

viii. $\text{Cos}\left(\text{Sin}^{-1}\frac{1}{\sqrt{2}}\right) =$..... (a) $\frac{2}{\sqrt{3}}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{1}{2}$ (d) $\frac{1}{\sqrt{2}}$

ix. $\cos^{-1}\text{A} + \cos^{-1}\text{B} =$ .....

(a) $\cos^{-1}\left(\text{AB} - \sqrt{(1-\text{A}^2)(1-\text{B}^2)}\right)$ (b) $\cos^{-1}\left(\text{AB} + \sqrt{(1-\text{A}^2)(1-\text{B}^2)}\right)$

(c) $\cos^{-1}\left(\text{AB} + \sqrt{(1+\text{A}^2)(1+\text{B}^2)}\right)$ (d) $\cos^{-1}\left(\text{AB} - \sqrt{(1+\text{A}^2)(1+\text{B}^2)}\right)$

x. $\tan^{-1}(-x) = \cdots$ (a) $-\tan^{-1}x$ (b) $\pi - \tan^{-1}x$ (c) $\cot^{-1}x$ (d) $\tan^{-1}x$

xi. $\text{Tan}^{-1}(-1) =$--------- (a) $\frac{\pi}{4}$ (b) $-\frac{\pi}{4}$ (c) $\frac{\pi}{2}$ (d) $-\frac{\pi}{2}$

xii. If $\tan 2x = -1$, then solution in the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is:

(a) $-\frac{\pi}{8}$ (b) $-\frac{\pi}{4}$ (c) $\frac{3\pi}{8}$ (d) $\frac{3\pi}{4}$

xiii. General solution of $4\sin x - 8 = 0$ is:

(a) $\{\pi + 2n\,\pi\}, n \in \mathbb{Z}$ (b) $\{\pi + n\,\pi\}, n \in \mathbb{Z}$

(c) $\{-\pi + n\,\pi\},\ n \in \mathbb{Z}$ (d) Not possible

xiv. Solution of equation: $2\,\text{Sin}\,x + \sqrt{3} = 0$ in 4th Quadrant is:

(a) $\frac{\pi}{3}$ (b) $\frac{-\pi}{3}$ (c) $\frac{-\pi}{6}$ (d) $\frac{11\pi}{6}$

xv. All trigonometric function are ..... function:

(a) periodic (b) even (c) injective (d) bijective

xvi. General solution of every trigonometric equation consists of:

(a) one solution only (b) two solution

(c) infinitely many solutions (d) no real solution

xvii. The period of $\cos\theta$ is:

(a) $\frac{\pi}{2}$ (b) π (c) 2π (d)4π

xviii. One solution of $\text{Sec}\,x = -2$ is:

(a) $\frac{2\pi}{3}$ (b) $\frac{\pi}{3}$ (c) $\frac{4\pi}{5}$ (d) $\frac{-\pi}{3}$

xix. If $\cos\theta = -\frac{1}{2}$ and $\sin\theta = \frac{-\sqrt{3}}{2}$, then θ is:

(a) $\frac{\pi}{3}$ (b) $\frac{2\pi}{3}$ (c) $\frac{4\pi}{3}$ (d) $\frac{5\pi}{3}$

xx. The period of $\sin\frac{x}{2}$ is

(a)4π (b)π (c) 2π (d) 3π

xxi. The value of $\cos 5\pi$ is

(a) 0 (b)1 (c) -1 (d) None of these

xxii. The period of $3\sec\frac{x}{3}$ is

(a)5π (b)π (c) 2π (d) 6π

xxiii. The period of $7\tan(7x)$ is

(a)$\frac{7\pi}{3}$ (b)7π (c) $\frac{\pi}{7}$ (d) π

2. Find the maximum and minimum value of the each of the following functions.

(i) $y = 5 - 7\cos\theta$ (ii) $y = 4 + 3\sin(2\theta - 5)$

3. By using graph, find the solution of the following equation.

$2\cos\theta - \theta = 0$

4. Without using calculator, show that:

(i) $\text{cosec}^{-1}\left(\frac{13}{12}\right) = \tan^{-1}\left(\frac{12}{15}\right)$ (ii) $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

5. Show that: $\cos^{-1}\left(\frac{7}{25}\right) - \cos^{-1}\left(\frac{5}{13}\right) = \cos^{-1}\left(\frac{323}{325}\right)$.

6. Find the solution sets of the equation $\sec 3x = \sec x$.